

1. We discussed the selection bias/survivorship bias in class. Can you give some other examples of this bias?
2. Explain the meaning of the OLS Assumption 1 by using diagrams. Give an example where $E[u] = 0$ but $E[u|X] \neq 0$ by using diagrams.
3. Suppose we want to find out if there exists difference in average wages between graduates from the School of Social Sciences (SSS) and graduates from the School of Humanities (SOH) using the linear regression model with an indicator variable ($X = 1$ or 0).
 - (a) Write down the linear regression model for this problem. Define the dependent variable, explanatory variable, and error term in your model.
 - (b) Write down the corresponding OLS Assumption 1. Explain in what case the OLS Assumption 1 is likely to hold/unlikely to hold.
 - (c) Now suppose that the OLS Assumption 1 does hold, explain the specific meaning of the parameters β_0 and β_1 using conditional expectation.
 - (d) Illustrate the meaning of β_0 and β_1 by using a diagram.
4. Let $\hat{\beta}_0$ and $\hat{\beta}_1$ be the estimated intercept and slope from the OLS regression of the linear regression model: $Y_i = \beta_0 + \beta_1 X_i + u_i$, $i = 1, \dots, n$. Let $\tilde{\beta}_0$ and $\tilde{\beta}_1$ be the estimated intercept and slope from the OLS regression of $\tilde{Y}_i$ on $\tilde{X}_i$, where $\tilde{Y}_i = 10 \cdot Y_i$ and $\tilde{X}_i = 10 \cdot X_i$, $i = 1, \dots, n$. Show the relation between the formula of $\tilde{\beta}_1$ and that of $\hat{\beta}_1$, and the relation between the formula of $\tilde{\beta}_0$ and that of $\hat{\beta}_0$.
(Hint: Write down the formulas of the OLS estimators $\tilde{\beta}_0$ and $\tilde{\beta}_1$.)